Worksheet 5- Chapters 5.1, 5.2 and 5.3

5.1 Introduction to Discrete probability distributions

5.2 The Binomial probability distribution.

5.1 An economics quiz contains six multiple choice questions. Let x represent the number of questions a student answer correctly.

a. Is x a continuous or discrete random variable?

b. What are the possible values of x?

5.22 The manager for state bank and trust has recently examined the credit card account balances for the customers of her bank and found that 20% have an outstanding balance at the credit card limit. Suppose the manager randomly selects 15 customers.

a. What is the probability that exactly 4 customers in the sample will have balances at the limit of the credit card? (Use the formula)

b. What is the probability that 4 or fewer customers in the sample will have balances at the limit of the credit card? (use the table)

5.31. Use the binomial distribution table to determine the following probabilities:

a. $n = 6$, $p = 0.08$; find $P(x = 2)$

b. $n = 11$, $p = 0.65$; calculate $P(2 < x \leq 5)$

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5.32. Use the binomial distribution in which n=6 and p = 0.3 to calculate the following probabilities:

a. x is at most 1.

b. x is at least 2.

c. x is more than 5.

d. x is less than 6.

5.34. Magic valley Memorial Hospital Administrators have recently received an internal audit report that indicates that patient bills contain an error of one form or another. After spending considerable efforts to improve the hospital's billing process, the administrators are convinced that things have improved. They believe that the new error rate is somewhere close to 0.05. (**self-study**)

a) what is the probability that they would find 3 or more errors in a sample of 10 bills? (use table)

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5.39. A survey by KRC Research for U.S News reported that 37% of people plan to spend more on eating out after they retire. If 8 people are randomly selected, then determine the

a. Expected number of people who plan to spend more on eating out after they retire.

b. Standard deviation of the individuals who plan to spend more on eating out after they retire.

c. Find the probability that two or fewer people in the sample actually plan to spend more on eating out after retirement**. (Use the formula**)

5.3 Other Probability Distributions

5.50. The mean number of errors per page made by a member of the word-processing pool for a large company is thought to be 1.5, with the number of errors distributed according to a Poisson distribution. If three pages are examined, what is the probability that more than three errors will be observed?

5-51. Arrivals to a bank automated teller machine (ATM) are distributed according to a Poisson distribution with a mean equal to three per 15 minutes.

a. Determine the probability that in a given 15-minute segment, no customers will arrive at the ATM.

b. What is the probability that fewer than four customers will arrive in a 30-minute segment?

5-54. If a random variable follows a Poisson distribution with $\lambda = 20$ and $t = \frac{1}{2}$ find the

a. expected value, variance, and standard deviation of this Poisson distribution.

b. probability of exactly 8 successes